ACADEMIUM ENGINE PACKET

Section 2: Signal Extraction and Competency Mapping

Version: 2.0 **Status:** Active Specification **Scope:** Applies to all Free Response (FR) decision inputs at Turns 2 and 3. **Does not apply to:** Multiple Choice inputs (Turn 1 signals are pre-authored via tradeoff signatures).

2.1 Purpose and Pedagogical Foundation

Signal extraction is the engine's core intelligence function. It is the mechanism by which a student's free-text response is transformed into structured evidence of reasoning quality. Without rigorous signal extraction, the engine cannot generate consequences that feel causally connected to the student's thinking, and the dashboard cannot produce defensible competency data.

Why This Matters

The entire value proposition of Academium rests on one claim: **it makes reasoning visible.** That claim is only true if the engine is actually detecting and measuring reasoning — not simulating the appearance of responsiveness while ignoring the content of what the student wrote.

Signal extraction is the mechanism that closes that gap.

Research Basis

Signal extraction is grounded in three converging research frameworks:

Bloom's Revised Taxonomy (Anderson & Krathwohl, 2001): Academium targets the upper three cognitive levels — *Apply*, *Analyze*, and *Evaluate*. Each signal type maps to one or more of these levels. The engine is not measuring whether students *remember* case facts; it is measuring whether they can *apply* them to a decision, *analyze* the implications, and *evaluate* the trade-offs.

Naturalistic Decision-Making (Klein, 1993, 2008): Professional decisions are made under uncertainty, with incomplete information, under time pressure. The signals the engine looks for (intent, justification, tradeoff awareness, stakeholder awareness, ethical awareness) are the exact cognitive moves that research identifies as markers of expert-level decision-making in naturalistic environments.

Feedback Intervention Theory (Kluger & DeNisi, 1996): Feedback that directs attention to the *task and reasoning process* improves learning. Feedback that evaluates the *person* impairs it. Signal extraction enables the engine to generate task-focused, reasoning-specific feedback —

because the engine knows what the student actually reasoned about, not just what they concluded.

2.2 The Five Decision Signals

Every FR student input is evaluated for the presence and quality of five decision signals. These signals are independent — a student may demonstrate some without others. The engine does not require all five; the presence and quality of any signals detected drives the downstream pipeline.

SIGNAL 1: INTENT

Definition: The student commits to a clear direction, priority, or course of action.

What this measures: Decision-making under uncertainty (Bloom's: *Apply*). The student has moved from observation to commitment. They are not describing the situation — they are engaging with it.

Detection criteria: The engine looks for a statement that names or implies a specific direction. Intent is present when the student has answered (explicitly or implicitly) the question: *"What would you do, and toward what end?"*

Quality Level	Criteria	Example
STRONG	Specific, directional, case-anchored. Names a concrete action or priority tied to a specific case element (a named stakeholder, a stated constraint, a defined timeline).	*"I would prioritize maintaining the client relationship by requesting a two-week extension, given that they represent 30% of projected Q3 revenue."*
PRESENT	Directional but generic. Names a direction without connecting it to specific case elements.	*"I would focus on preserving the client relationship."*
WEAK	Hedged or implied. The student suggests a direction but does not commit to it. Language includes: "maybe," "perhaps," "could consider," "might be worth," "I think it would be good to."	*"Maybe we should try to work something out with the client."*
ABSENT	No directional statement. The student describes, observes, or questions the situation without committing to a course of action.	*"This is a difficult situation with many factors to consider."*

Common false positives:

- A question mistaken for intent: *"Shouldn't we talk to the client?"* — This is a prompt for discussion, not a decision commitment.
- A restated case fact: *"The client needs a faster timeline"* — This is description, not intent.

SIGNAL 2: JUSTIFICATION

Definition: The student provides reasoning for their position — a causal explanation for why their chosen direction is appropriate given the situation.

What this measures: Analytical thinking and causal reasoning (Bloom's: *Analyze*). The student is not just stating what they would do — they are explaining the logic that connects the situation to the action.

Detection criteria: Justification is present when the student's response contains a reasoning chain, however brief. Linguistically, look for causal connectors: *because, since, given that, therefore, as a result of, which means, this would allow, otherwise.* But justification can also be implicit — a conditional structure ("if X, then Y") or a consequence structure ("doing X avoids Y") both qualify.

Quality Level	Criteria	Example
STRONG	Case-specific causal chain. The reasoning explicitly connects a case element (data point, stakeholder position, constraint) to the decision.	*"Because the operations team has already committed resources to Q3 delivery, renegotiating now avoids a more costly disruption later."*
PRESENT	General causal reasoning. The student provides a logical reason, but it could apply to many situations and is not anchored to this specific case.	*"Because maintaining trust with clients is fundamental to long-term business success."*
WEAK	Asserted without reasoning. The student states a conclusion with minimal or circular explanation. "Because it's the right thing to do." "Because it's important." "Because it's better."	*"Because we need to handle this properly."*
ABSENT	No reasoning provided. The student states a position with no explanation.	*"I would talk to the client."*

Common false positives:

- Description mistaken for justification: *"The budget has been cut by 15%"* — This is a case fact restatement, not a reasoning step.
- Tautology: *"We should prioritize the client because the client is our priority"* — Circular; does not advance reasoning.

SIGNAL 3: TRADEOFF AWARENESS

Definition: The student explicitly or implicitly acknowledges what they are giving up, risking, or sacrificing by choosing their stated direction.

What this measures: Strategic thinking and multi-criteria evaluation (Bloom's: *Evaluate*). This is one of the most important signals in the engine because it directly measures the student's capacity to think beyond a single dimension — a core competency in business decision-making.

Important: Tradeoff Awareness is always detected in student responses regardless of whether the professor has configured tradeoffs for this case. The professor's tradeoff configuration governs consequence generation behavior (Section 3), not signal detection. A student who demonstrates tradeoff awareness always gets credit for that reasoning — the engine always records it.

Detection criteria: Tradeoff awareness is present when the student's response acknowledges a cost, risk, side effect, or sacrifice associated with their chosen direction. This does not require formal tradeoff language — it can be implied by mentioning a competing priority, a stakeholder who will be negatively affected, or a constraint that will be strained.

Quality Level	Criteria	Example
STRONG	Specific, named tradeoff with consequence. Identifies a specific cost and connects it to a specific case element or stakeholder.	*"This will likely delay the internal project timeline by at least two weeks, which means the operations team will need to defer their Q3 efficiency targets."*
PRESENT	Acknowledged tradeoff without specifics. The student recognizes there is a cost but does not name it concretely.	*"This might create some internal pressure on the team."*
WEAK	Generic acknowledgment. A formulaic statement that tradeoffs exist without any specificity.	*"There are always pros and cons to every decision."*
ABSENT	No acknowledgment of cost or sacrifice. The student presents their direction as having no meaningful downside.	*(A response that only describes benefits of the chosen direction with no acknowledgment of what it costs.)*

Important note on tradeoff quality: A student who identifies a tradeoff that is *factually wrong* (misreads the case) has still demonstrated tradeoff *awareness* as a cognitive move, even if the specific tradeoff is incorrect. The engine should credit the presence of the cognitive behavior. The consequence narrative can correct the factual misread through the simulation dynamics.

SIGNAL 4: STAKEHOLDER AWARENESS

Definition: The student considers the impact of their decision on a specific person, team, group, or organizational entity beyond the immediate decision itself.

What this measures: Stakeholder management and systems thinking (Bloom's: *Analyze/Evaluate*). Research on professional competency consistently identifies stakeholder awareness as a key differentiator between novice and experienced decision-makers. Novices optimize for a single outcome; experienced professionals map how their decisions ripple through the people and systems around them.

Detection criteria: Stakeholder awareness is present when the student names or clearly implies a specific group and connects them to the decision's impact. Generic references ("the company," "everyone," "the people involved") do not qualify. Named or clearly implied groups do ("the operations team," "our client," "the board," "junior staff").

Quality Level	Criteria	Example
STRONG	Named stakeholder with specific impact. Identifies who is affected, how they are affected, and why that matters for the decision.	*"The operations team will need to adjust their resource allocation, and we should loop them in before making any commitment to the client to avoid setting expectations we can't meet."*
PRESENT	Named stakeholder, general impact. Mentions a specific group but describes impact vaguely.	*"The operations team would be affected by this decision."*
WEAK	Implied stakeholder, no specificity. References "the team," "employees," "stakeholders" without identifying who specifically or how.	*"The team would need to adapt."*
ABSENT	No stakeholder consideration beyond the immediate actor. The student discusses the decision as if it affects only themselves or the abstract organization.	*(A response focused entirely on financial metrics with no mention of human impact.)*

Common false positives:

- The student as stakeholder: *"I would need to think carefully about this"* — This is not stakeholder awareness; it is self-reference.
- The client/customer as the only stakeholder, when the case clearly involves internal stakeholders too — this is PRESENT at best, because the student is missing part of the stakeholder picture.

SIGNAL 5: ETHICAL AWARENESS

Definition: The student surfaces a principle, obligation, fairness concern, transparency requirement, or moral dimension relevant to the decision.

What this measures: Ethical reasoning (Bloom's: *Evaluate*). Note: this signal is not about whether the student makes the "ethical" choice. It is about whether they *recognize* that an ethical dimension exists in the situation and incorporate it into their reasoning.

Detection criteria: Ethical awareness is present when the student invokes a concept of obligation, fairness, transparency, responsibility, or harm — and connects it to the case situation.

The ethical dimension must be applied, not abstract.

Quality Level	Criteria	Example
STRONG	Applied ethical principle with case connection. Names a specific ethical obligation or concern and connects it to a specific stakeholder or outcome in the case.	*"We have an obligation to be transparent with the client about the delay, even if it risks the relationship — they've made commitments to their own stakeholders based on our timeline."*
PRESENT	Ethical principle acknowledged but generic. Invokes an ethical concept without fully applying it to the specific case.	*"We should be honest with the client about what's happening."*
WEAK	Abstract ethical language without application. Invokes broad moral language that doesn't connect to the specific situation.	*"We need to do the right thing here."*
ABSENT	No ethical dimension surfaced. The student's response is entirely consequentialist (focused on outcomes/metrics) or technical (focused on process/execution) with no acknowledgment of obligation or responsibility.	*(A response focused only on financial recovery strategy with no mention of transparency or stakeholder obligations.)*

Important clarification: Ethical awareness does NOT require the student to take an "ethical" position. A student who says *"The financial pressure makes it tempting to delay informing the client, but we have a transparency obligation"* is demonstrating strong ethical awareness — even though they've identified a tension, not just a clean ethical answer. Recognizing the tension IS the ethical reasoning.

2.3 Signal Quality Scoring

After signal extraction, the engine computes a **Reasoning Depth Score (RDS)** for the input. The RDS is an internal value used to govern consequence richness and KPI tier selection. It is **never displayed to the student, never used as a grade, and never communicated to the professor as a performance metric.** It is a pipeline variable only.

RDS Computation

Each signal is scored as follows:

Quality Level	Score
STRONG	3 points
PRESENT	2 points
WEAK	1 point
ABSENT	0 points

Maximum RDS = 15 (all five signals at STRONG) **PASS threshold = any signal scoring ⩾ 2 (PRESENT or above)**

RDS Bands and Pipeline Effects

RDS Band	Range	Pipeline Effect
Surface	1–4	Student has cleared PASS but reasoning is thin. Consequence narrative is shorter and more direct. KPI movements are Tier 1 only. Causal explanations are simpler.
Engaged	5–9	Student demonstrates genuine engagement with the decision. Standard consequence narrative length. Tier 1 and Tier 2 KPI movements possible. Causal explanations are specific and reference student signals.
Integrated	10–15	Student demonstrates multi-signal reasoning. Richer consequence narrative with compounded effects visible. Tier 2 movements standard; Tier 3 possible if pre-authored in case. Causal explanations directly quote or paraphrase student reasoning.

Critical Rules

- **RDS never triggers a NUDGE or BLOCK.** Those are governed by the PASS/NUDGE/BLOCK classification system in Section 6 of the Constitution. A low RDS means thin consequences — it does not mean the input is rejected.
- **A two-word PASS with a single PRESENT-level signal gets an RDS of 2.** The simulation proceeds. The student gets Surface-tier consequences. This is correct behavior.
- **Grammar, spelling, and length are never scoring criteria.** A grammatically imperfect response with a strong causal reasoning chain scores higher than a polished but empty

response.

2.4 Competency Inference Logic

Signal data feeds the competency inference layer. This layer translates extracted signals into competency evidence — the structured data that populates the dashboard and supports AACSB Assurance of Learning reporting.

The Five Academium Competencies

These five competencies are Academium's canonical measurement targets, derived from AACSB learning goals and aligned with the business education research base.

Competency ID	Competency Name	Definition
C1	Analytical Reasoning	The ability to identify relevant information, construct logical arguments, and support conclusions with evidence
C2	Strategic Decision-Making	The ability to make directional commitments under uncertainty, weighing constraints and consequences
C3	Stakeholder Consideration	The ability to identify and account for the interests and impacts across multiple affected parties
C4	Ethical Reasoning	The ability to recognize ethical dimensions of decisions and incorporate principles of responsibility, fairness, and transparency
C5	Systems Awareness	The ability to recognize how decisions create second-order effects across interconnected organizational, financial, and human systems

Signal-to-Competency Mapping

A single signal can contribute evidence to one or more competencies. The following table defines the mapping:

Signal	Primary Competency	Secondary Competency	Notes
INTENT	C2 (Strategic Decision-Making)	—	Intent alone demonstrates C2 only at baseline. Strong intent with anchoring to case constraints also contributes to C1.
JUSTIFICATION	C1 (Analytical Reasoning)	C2 (Strategic Decision-Making)	Justification is the primary evidence source for C1. Strong justification with case-specific data elevates C2 evidence quality.
TRADEOFF AWARENESS	C5 (Systems Awareness)	C2 (Strategic Decision-Making)	Tradeoff awareness is the primary evidence source for C5. Named, specific tradeoffs also contribute to C2 evidence.
STAKEHOLDER AWARENESS	C3 (Stakeholder Consideration)	C5 (Systems Awareness)	Stakeholder awareness is the primary and near-exclusive evidence source for C3. Multi-stakeholder responses also contribute to C5.
ETHICAL AWARENESS	C4 (Ethical Reasoning)	C3 (Stakeholder Consideration)	Ethical awareness is the primary evidence source for C4. Applied ethical reasoning (not just abstract) also contributes to C3 when it names affected parties.

Competency Evidence Levels

For dashboard purposes, each competency is reported at one of three evidence levels per decision point:

Evidence Level	Criteria
Demonstrated	The mapped signal is PRESENT or STRONG quality. The student's response contains clear evidence of this competency.
Emerging	The mapped signal is WEAK quality. The student shows awareness of the competency but has not yet applied it with specificity.
Not Evidenced	The mapped signal is ABSENT. This decision point produced no evidence for this competency.

Critical rule: "Not Evidenced" does not mean "student lacks this competency." It means this decision point did not produce evidence. Competency assessment requires pattern evidence across multiple decision points, not a single data instance.

2.5 AACSB Alignment Map

Academium's five competencies map to AACSB's standard learning goals as follows:

Academium Competency	AACSB Learning Goal Alignment	Evidence Type
C1: Analytical Reasoning	Analytical and Critical Thinking	Direct (decision log with justification trace)
C2: Strategic Decision-Making	Application of Knowledge; Reflective Thinking	Direct (decision commitment under uncertainty, consequence response)
C3: Stakeholder Consideration	Diverse Perspectives; Communication	Direct (named stakeholder impacts in decision log)
C4: Ethical Reasoning	Ethical Understanding and Reasoning	Direct (ethical dimension identification in decision log)
C5: Systems Awareness	Analytical and Critical Thinking; Reflective Thinking	Direct (tradeoff and second-order effect identification)

How This Supports AoL Reporting

Every Academium session generates a **Decision Evidence Log** — a structured record of the

student's inputs, extracted signals, competency evidence levels, and decision outcomes. This log is the raw evidence for AoL reporting.

The professor maps their course learning objectives to Academium's five competencies when configuring a case. The dashboard aggregates individual Decision Evidence Logs to produce:

- **Cohort competency heat maps** (where is the class strong? where are the gaps?)
- **Individual trajectory reports** (how did this student's reasoning evolve across the simulation?)
- **AoL evidence packages** (formatted for accreditation reporting, mapping student performance to stated learning goals)

2.6 Prohibited Inferences

The following inferences are explicitly prohibited. The engine must never make these translations, regardless of input quality:

Prohibited Inference	Why It Is Prohibited
High RDS = correct decision	RDS measures reasoning quality, not decision correctness. There is no correct decision.
Low RDS = student is weak	A single decision point with low signal density is insufficient basis for a capability judgment.
Absent ethical signal = student is unethical	The student may hold strong ethical values and simply not have surfaced them in this response. Signal absence is a data point about this response, not a judgment about the person.
Strong intent + absent justification = good strategic thinking	Intent without reasoning is directional, not strategic. The engine must not infer reasoning quality from directional commitment alone.
Keyword presence = signal presence	The engine must detect semantic meaning, not surface keywords. "I think about stakeholders a lot" does not constitute Stakeholder Awareness. "The operations team will need to reallocate their Q3 resources" does.

2.7 Signal Extraction Worked Examples

The following examples demonstrate correct signal extraction across the quality spectrum. All examples are in Spanish (Case 001 language). English translations are provided in brackets.

EXAMPLE A — High RDS Response (Integrated Band)

Student input: *"Priorizaría renegociar el plazo con el cliente, siendo transparentes sobre la reducción presupuestaria. Si bien esto puede generar tensión inicial, perder la confianza del cliente costaría más a largo plazo que el retraso mismo. El equipo de operaciones tendrá que ajustar sus compromisos de Q3, algo que debemos coordinar antes de confirmar cualquier plazo al cliente."*

["I would prioritize renegotiating the timeline with the client, being transparent about the budget reduction. While this may create initial tension, losing the client's trust would cost more in the long run than the delay itself. The operations team will need to adjust their Q3 commitments, something we must coordinate before confirming any timeline to the client."]

Signal extraction:

Signal	Quality	Evidence
Intent	STRONG	"Priorizaría renegociar el plazo con el cliente" — specific, directional, case-anchored
Justification	STRONG	Causal chain: transparency → initial tension but → long-term trust cost exceeds delay cost
Tradeoff Awareness	STRONG	"puede generar tensión inicial" + Q3 commitment adjustment — names specific costs
Stakeholder Awareness	STRONG	Operations team named with specific impact (Q3 adjustment); client named with trust consequence
Ethical Awareness	PRESENT	Transparency invoked ("siendo transparentes") — applied to situation but not elaborated as obligation

RDS: 3+3+3+3+2 = **14 (Integrated Band) Competency evidence:** C1 Demonstrated, C2 Demonstrated, C3 Demonstrated, C4 Demonstrated, C5 Demonstrated **Pipeline effect:** Rich consequence narrative, compounded effects, Tier 2 KPI movements eligible

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EXAMPLE B — Mid RDS Response (Engaged Band)

Student input: *"Creo que lo mejor sería hablar con el cliente y explicar la situación. No podemos ignorar el problema y el cliente merece saber qué está pasando. Esto puede afectar al equipo internamente, pero mantener la relación es más importante."*

["I think the best thing would be to talk to the client and explain the situation. We can't ignore the problem and the client deserves to know what is happening. This may affect the team internally, but maintaining the relationship is more important."]

Signal extraction:

Signal	Quality	Evidence
Intent	PRESENT	"hablar con el cliente y explicar la situación" — directional but generic (no specific case anchoring)
Justification	PRESENT	"mantener la relación es más importante" — general causal reasoning, not case-specific
Tradeoff Awareness	PRESENT	"puede afectar al equipo internamente" — acknowledged but vague
Stakeholder Awareness	PRESENT	Client named; team implied but not specified
Ethical Awareness	PRESENT	"el cliente merece saber" — fairness/transparency implied, applied to situation

RDS: 2+2+2+2+2 = **10 (Engaged Band) Competency evidence:** C1 Emerging→Demonstrated, C2 Demonstrated, C3 Demonstrated, C4 Demonstrated, C5 Emerging **Pipeline effect:** Standard consequence narrative, Tier 1-2 KPI movements, causal explanations reference student reasoning

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EXAMPLE C — Low RDS Response (Surface Band, still PASS)

Student input: *"Hablaría con el cliente."*

["I would talk to the client."]

Signal extraction:

Signal	Quality	Evidence
Intent	PRESENT	"Hablaría con el cliente" — clear, directional commitment. Short but unambiguous.
Justification	ABSENT	No reasoning provided
Tradeoff Awareness	ABSENT	No acknowledgment of cost
Stakeholder Awareness	WEAK	Client implied; no other stakeholders
Ethical Awareness	ABSENT	No ethical dimension surfaced

RDS: 2+0+0+1+0 = **3 (Surface Band) PASS/NUDGE/BLOCK classification:** PASS (clear priority stated — "Hablaría con el cliente") **Note:** Word count alone does NOT trigger NUDGE. This is a valid PASS with thin reasoning. The engine delivers Surface-tier consequences. The NUDGE/PASS logic is in Section 6 of the Constitution. **Competency evidence:** C2 Demonstrated (intent), all others Not Evidenced for this decision point

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EXAMPLE D — NUDGE-triggering Response

Student input: *"Es una situación complicada y hay que pensar bien qué hacer."*

["It's a complicated situation and you have to think carefully about what to do."]

Signal extraction:

Signal	Quality	Evidence
Intent	ABSENT	No directional commitment
Justification	ABSENT	No reasoning
Tradeoff Awareness	ABSENT	None
Stakeholder Awareness	ABSENT	None
Ethical Awareness	ABSENT	None

RDS: 0 — but NOTE: RDS does not determine PASS/NUDGE/BLOCK **PASS/NUDGE/BLOCK classification:** NUDGE (relevant to the case, shows engagement — "situación complicada" — but

no extractable decision signals) **Engine action:** Preserve student text. Display non-blocking mentor nudge. Suggest one clarifying question: *"Tu observación es válida. Para poder avanzar, ¿qué acción concreta tomarías y por qué?"*

2.8 Section Summary: What the Engine Must Do

At every FR decision point, the engine must:

1. **Extract all five signals** independently and assign a quality level (STRONG / PRESENT / WEAK / ABSENT) to each.
2. **Compute the RDS** from the quality level scores.
3. **Map detected signals to competencies** using the Signal-to-Competency table.
4. **Assign competency evidence levels** (Demonstrated / Emerging / Not Evidenced) for each of the five competencies.
5. **Write the RDS band and competency evidence** to the Decision Evidence Log for dashboard use.
6. **Pass the RDS band to the Consequence Generation step** to govern narrative richness and KPI tier selection.
7. **Never** infer student character, capability, or performance from any single decision point.
8. **Never** display the RDS, signal scores, or competency evidence to the student.
9. **Never** allow signal absence to trigger a BLOCK or punitive consequence narrative.

Section 2 connects to: Section 3 (Consequence Generation — receives RDS band), Section 4 (KPI Computation — receives signal data for metric direction), Section 5 (Causal Explanation Generation — receives specific signal content for student-referenced explanations), Section 7 (Dashboard — receives Decision Evidence Log).